Math Thermo
class #12
02/26/2026

The temperature relation can be inverted for U, i.e., there is a function $\tilde{U} = \tilde{U}(T,N)$, such that

$$\tilde{U}(T(U,N),N) = U, \quad \forall (U,N) \in \mathrm{dom}(T)$$

Thus, there is a function

$$\tilde{S}_{eq} = \tilde{S}_{eq}(T,N)$$

defined by

(12.1) $$\tilde{S}_{eq}(T,N) := S_{eq}(\tilde{U}(T,N),N).$$

<u>Definition</u> (12.1): The statistical Helmholtz free energy is defined as

(12.2) $$F(T,N) := \tilde{U}(T,N) - T\tilde{S}_{eq}(T,N),$$

for all $(T,N) \in \mathrm{dom}(\tilde{U})$.

<u>Theorem</u> (12.2): For the present case (N non-interacting particles with internal energy U), it follows that

(12.3) $$F(T,N) = -N k_B T \log(Z),$$

where the partition function satisfies

(12.4)
$$Z = P_0 \iint_Q \exp\left(-\frac{u(\vec{q})}{k_B T}\right) d\vec{q}.$$

Similar to before,

(12.5)
$$\tilde{U}(T,N) = N k_B T^2 \frac{\partial}{\partial T}\left[\log(Z)\right]$$

and

(12.6)
$$\tilde{S}_{eq}(T,N) = N k_B \log(Z) + N k_B T \frac{\partial}{\partial T}\left[\log(Z)\right],$$

Proof: The proofs are similar to those for the discrete energy case.

The details are left for an exercise. ///

Ideal Gas

For a monotomic ideal gas, we take the per-particle energy to be

$$u(\vec{q}) = \frac{1}{2} m (v_1^2 + v_2^2 + v_3^2)$$

for particles in the cube $\Omega = (0,L)^3$. The volume is of course

$$V = \int_\Omega d\vec{x} = L^3.$$

The partition function determines the thermodynamic variables in the system.

$$Z = P_0 \iint_{\mathbb{R}^3\,\Omega} \exp\left(\frac{-m|\vec{v}|^2}{2k_BT}\right) d\vec{x}\, d\vec{v}$$

$$= P_0 V \left[\int_{\mathbb{R}} \exp\left(\frac{-mv^2}{2k_BT}\right) dv\right]^3$$

(12.7) $$= P_0 V \left(\frac{2\pi k_B T}{m}\right)^{3/2}.$$

Now, we need to compute $\log(Z)$. Recall that

$$[Z] = 1.$$

Let $T_0 > 0$ be a reference temperature. Then define

(12.8) $$v_0 := \sqrt{\frac{2\pi k_B T_0}{m}},$$

which has units of velocity. Then

$$Z = P_0 V \left(\frac{2\pi k_B T_0}{m}\right)^{3/2} \left(\frac{T}{T_0}\right)^{3/2}$$

(12.9) $$= P_0 V v_0^3 \left(\frac{T}{T_0}\right)^{3/2}.$$

Note

$$[P_0 V v_0^3] = 1 = \left(\frac{T}{T_0}\right)^{3/2}$$

Thus,

$$\log(Z) = \log(P_0 V v_0^3) + \frac{3}{2}\log\left(\frac{T}{T_0}\right). \tag{12.10}$$

Here, we have been careful to make the arguments of the log dimensionless. Clearly,

$$\frac{\partial}{\partial T}\log(Z) = \frac{3}{2}\frac{T_0}{T}\cdot\frac{1}{T_0} = \frac{3}{2T}. \tag{12.11}$$

It follows that

$$\begin{aligned} F &= -Nk_BT\log(Z) \\ &= -Nk_BT\log(P_0 V v_0^3) \\ &\quad -\frac{3Nk_BT}{2}\log\left(\frac{T}{T_0}\right), \end{aligned} \tag{12.12}$$

$$\begin{aligned} U &= Nk_BT^2\frac{\partial}{\partial T}\left[\log(Z)\right] \\ &= \frac{3Nk_BT}{2}, \qquad \text{(linear in } T\text{)} \end{aligned} \tag{12.13}$$

and

$$\begin{aligned} S &= Nk_B\log(Z) + Nk_BT\frac{\partial}{\partial T}\left[\log(Z)\right] \\ &= Nk_B\left\{\log(P_0 V v_0^3) + \frac{3}{2}\log\left(\frac{T}{T_0}\right)\right\} \\ &\quad + \frac{3Nk_B}{2}. \end{aligned} \tag{12.14}$$

For simplicity, we have dropped tildes, et cetera.

We can compute the pressure via

$$P = -\frac{\partial F}{\partial V}.$$

Recall that, in general, we have

$$dF = -S\,dT - P\,dV + \mu\,dN.$$

Thus,

$$P = \frac{N k_B T}{V}.$$

It follows that

(12.15) $$\boxed{PV = N k_B T.}$$

This is the familiar equation of state for an ideal gas.

Let us examine the Maxwell-Boltzmann distribution for the ideal gas. Recall,

$$f(\vec{x},\vec{v}) = f^{eq}(\vec{x},\vec{v}) = P_0 \frac{N}{Z} \exp\left(\frac{-m|\vec{v}|^2}{2k_B T}\right),$$

where

$$Z = P_0 V \left(\frac{2\pi k_B T}{m}\right)^{3/2} = P_0 V v_0^3 \left(\frac{T}{T_0}\right)^{3/2}.$$

Thus,

$$\boxed{f^{eq}(\vec{x},\vec{v}) = M(\vec{v}) = \frac{N}{V\left(\frac{2\pi k_B T}{m}\right)^{3/2}} \exp\left(\frac{-m|\vec{v}|^2}{2k_B T}\right).}$$

The function M is called the <u>Maxwellian</u>.

What is the average velocity of particles in our ideal gas?

What would you guess to be the answer?

$$\vec{v}_{ave} := \iint_Q \frac{f^{eq}(\vec{q})}{N} \vec{v}\, d\vec{q}$$

$$= \int_{\mathbb{R}^3} \frac{\exp\left(\frac{-m|\vec{v}|^2}{2k_B T}\right)}{\left(\frac{2\pi k_B T}{m}\right)^{3/2}} \vec{v}\, d\vec{v}$$

$$= \vec{0}.$$

This makes sense, right? We'll come back to this calculation a little later.

Next, let us show that, as expected, the temperature is positive.

$$\frac{3Nk_B T}{2} = U$$

$$= \iint_Q f^{eq}(\vec{q})\, u(\vec{q})\, d\vec{q}$$

$$= \iint_Q \overbrace{M(\vec{v})\,\frac{1}{2} m |\vec{v}|^2}^{\geq 0}\, d\vec{x}\, d\vec{v}$$

$$\geq 0$$

Thus,

$$T \geq 0.$$

The Regular Solution Model

Suppose two types of atoms, black and white,

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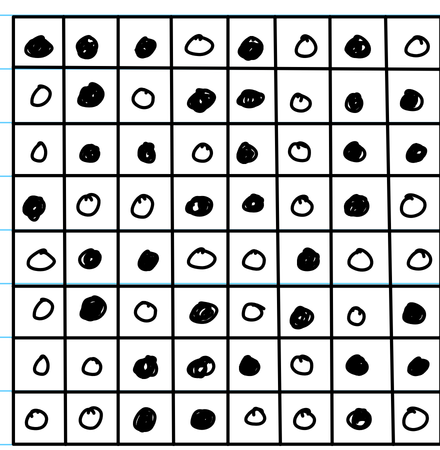
coexist on a cubic lattice:



N = # of lattice sites

N_b = # black atoms

N_w = # white atoms

There are 3 nearest neighbor energy levels

u_{bb} ● ●

u_{bw} ● ○

u_{ww} ○ ○

Each atom can sit only at a lattice point and all lattice points are occupied. Thus,

$$N = N_b + N_w$$

is fixed. Each atom occupies exactly $v_a > 0$ units of volume. Thus,

$$V = v_a N$$

is the total volume of the lattice in three dimensions.

For the system we have a finite number of discrete energy states

$$U_i = u_{bb} N_i^{bb} + u_{bw} N_i^{bw} + u_{ww} N_i^{ww}$$

where

$$N_i^{bb} = \# \text{ of pairs } \bullet\,\bullet$$

$$N_i^{bw} = \# \text{ of pairs } \circ\,\bullet$$

$$N_i^{ww} = \# \text{ of pairs } \circ\,\circ$$

This is an interacting particle system.

The partition function is

$$Z = \sum_i \exp\left(\frac{-U_i}{k_B T}\right).$$

Set

$$\phi := \frac{N_b}{N}.$$

Then

$$\frac{N_w}{N} = \frac{N - N_b}{N} = 1 - \phi.$$

The parameter ϕ is called the <u>number fraction of (black) atoms</u>.

<u>Mean-Field Approximation</u>

Let us approximate Z by

$$Z = \sum_i \exp\left(\frac{-U_i}{k_B T}\right)$$

$$\approx \sum_i \exp\left(\frac{-\bar{U}}{k_B T}\right)$$

$$= W \exp\left(\frac{-\bar{U}}{k_B T}\right)$$

where $\bar{U}$ is the average value of U and Ω the number of ways of placing N_b atoms in N cells. Thus,

$$W = \frac{N!}{N_b!\, N_w!}$$

Let us now estimate the mean (average) energy.

We will assume that N is very large, which allows us to neglect boundary effects.

The number of nearest neighbors, z, of any cell is the same!

in 3D $z = 6$ in 2D $z = 4$

If the atom fraction ϕ is known and we fix a lattice point, then, on average

- $z\phi$ near neighbor cells are occupied by black atoms,
- $z(1-\phi)$ near neighbor cells are occupied by white atoms.

Therefore,

$$\overline{N_{bb}} \approx \frac{1}{2} z \phi N_b = \frac{z}{2} N \phi^2 .$$

Similarly,

$$\overline{N_{ww}} \approx \frac{z}{2} N (1-\phi)^2 ,$$

and

$$\overline{N_{bw}} \approx z N \phi (1-\phi) .$$

Thus

$$\overline{U} \approx U_{bb} \overline{N_{bb}} + U_{ww} \overline{N_{ww}} + U_{bw} \overline{N_{bw}}$$

$$\approx \frac{z}{2} N \Delta U \phi^2 + N \left(C_0 + C_1 \phi \right)$$

where

$$\Delta U = U_{bb} + U_{ww} - 2 U_{bw} .$$

Recall that

$$F \cong -k_B T \log(Z) = -T S_{mix} + \overline{U}$$

where

$$S_{mix} = k_B \log(W) .$$

S_{mix} is called the entropy of mixing. Using the expression for W derived earlier,

$$S_{mix} = k_B \log(W)$$

$$= k_B \log(N!) - \log(N_b!) - \log(N_w!).$$

Next we use Stirling's approximation

$$\log(N!) \approx N\log(N) - N.$$

Thus,

$$S_{mix} = k_B\left[-N_b \log\left(\frac{N_b}{N}\right) - N_w \log\left(\frac{N_w}{N}\right)\right)$$

$$= k_B N\left[-\phi\log(\phi) - (1-\phi)\log(1-\phi)\right].$$

Hence

$$F \approx N k_B T\left\{\phi\log(\phi) + (1-\phi)\log(1-\phi)\right\}$$

$$+ N\frac{z}{2}\Delta U\phi^2 + N\left(C_0 + C_1\phi\right)$$

Recall

$$F = V f(T, \phi),$$

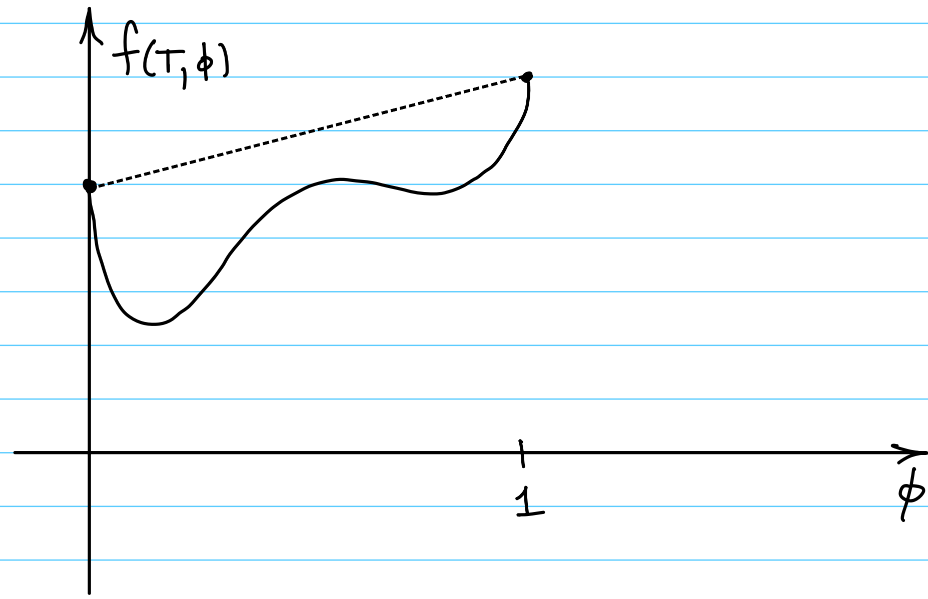
where f is the free energy density. Thus

$$f(T,\phi) = \frac{k_B T}{v_a}\left\{\phi\log(\phi) + (1-\phi)\log(\phi)\right.$$

$$\left. + \chi_2\phi(1-\phi) + \chi_1\phi\right\}$$

where

$$\chi_2 = -\frac{z}{2k_BT}\Delta U$$



Suppose that $\chi_1 = 0$ (symmetric case). There is a critical value of χ_2, called $\chi_{2,c}$, such that if $\chi_2 < \chi_{2,c}$, $f(T,\phi)$ is convex and if $\chi_2 > \chi_{2,c}$, $f(T,\phi)$ is non-convex.